

Team Members: Wyatt Victory, Nicolas Suaste

Faculty advisors: Dr. Vivek Khare

Dr. Rani Warsi Sullivan

Development of 3D Printed Compliant Hinges for Use in Thick Panel Origami Structures

Problem Statement

With growing interest in space travel, there is a need for lightweight, compact space structures to accomplish various missions. By leveraging certain material properties, loadbearing structures can be designed using compliant mechanisms and origami techniques to create bistable folding structures that save space without compromising functionality. The goal is to find 3D-printable materials suitable for this purpose and perform computational analysis and various stress tests to characterize the materials based on folding angles, fatigue, and optimal hinge thickness.

Overall Project Objective(s)

- Find 3D-printable materials suitable for compliant mechanisms
- Design a compliant hinge for use in thick panel origami structures
- Model and perform computational analysis on the hinge mechanism
- Characterize the materials based on performance
 - Fatigue testing
 - Optimal folding angles
 - Hinge thickness
 - Strength in bistable positions

Individual Team Member Responsibilities

Wyatt Victory: Material search, preliminary hinge design, modeling, computational analysis

Nicolas Suaste: Material search, experiment design, thickness optimization, computational analysis

Reporting

Reports will be provided weekly/bi-weekly in the form of memos detailing the work completed, issues encountered, and future plans. A designated time each week for each advisor will be found and agreed upon where in-person meetings can be requested by either party to further discuss the project and its current state.

Signatures

Those parties who willfully present their signatures on this document attest that they have read and understand all of the requirements and expectations presented above both explicitly and implicitly. This document represents a promise by the students to perform the task outlined above to the best of their ability. It is the responsibility of the course instructor and faculty advisor to understand the problem description outlined above and to require the students to abide by this statement of work.

List of team members and advisors with dated signatures.